

Numerical Analysis

Numerical Integration

A Resonance-free Adaptive Levin Method in Two-Dimensions

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$$\int_R f(\mathbf{x}) \exp(ig(\mathbf{x})) d\mathbf{x}$$

Adaptive Integration

- Deploy Levin Method
- Outperforms all methods in high frequency regime
- Time complexity is independent of frequency

ODE Solvers

A SOLVER FOR LINEAR SCALAR ORDINARY DIFFERENTIAL EQUATIONS
WHOSE RUNNING TIME IS BOUNDED INDEPENDENT OF FREQUENCY

MURDOCK AUBRY* AND JAMES BREMER†

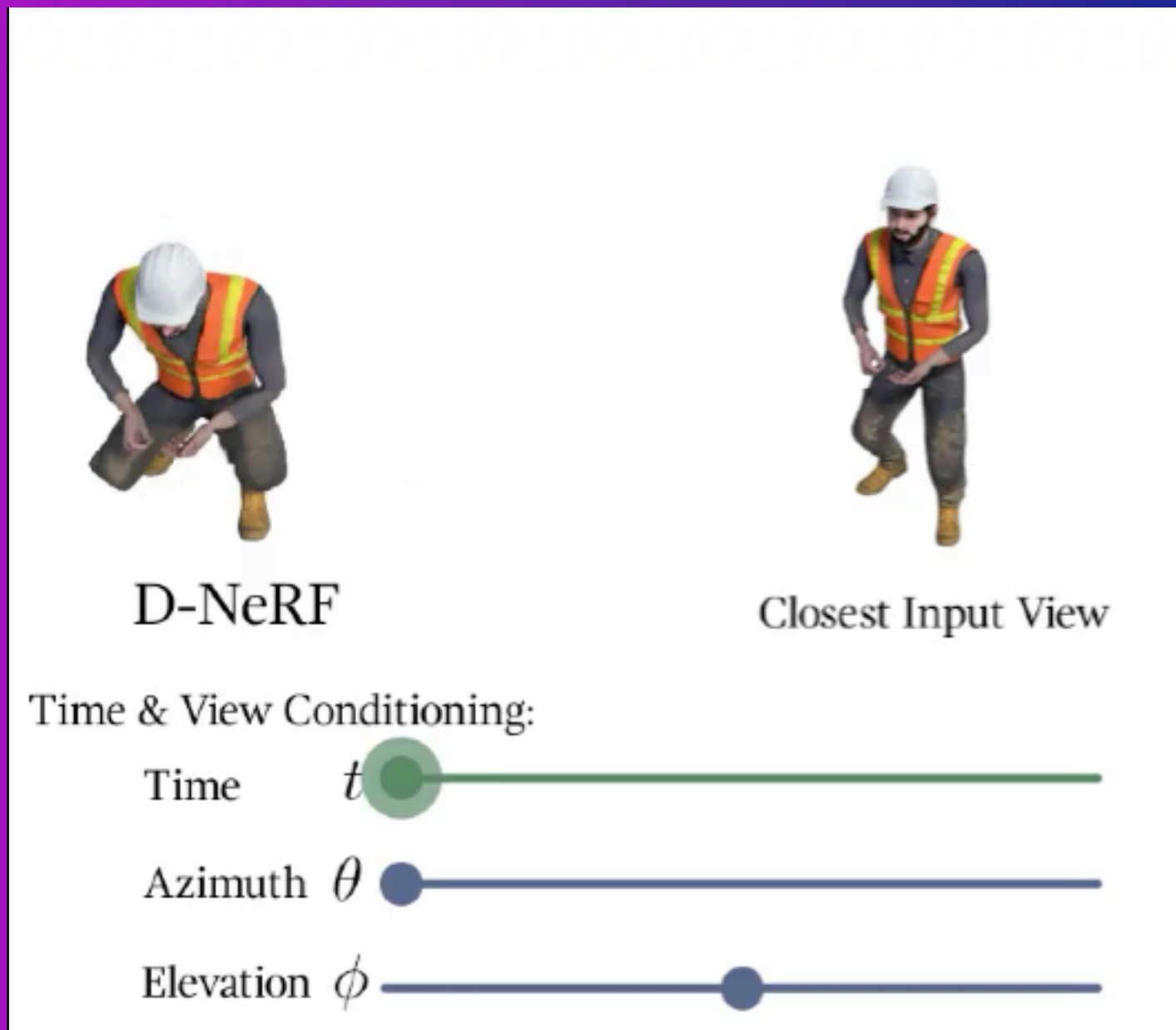
$$p(t)y(t) + \sum_{i=1}^N q_i(t) \frac{\partial^i y}{\partial t^i} = 0$$

Numerical solver for large order scalar equations

- Generalizes to all dimensions
- Outperforms all methods in high frequency regime
- Time complexity is independent of frequency

Physics-based Learning: NeRF

D-NeRF

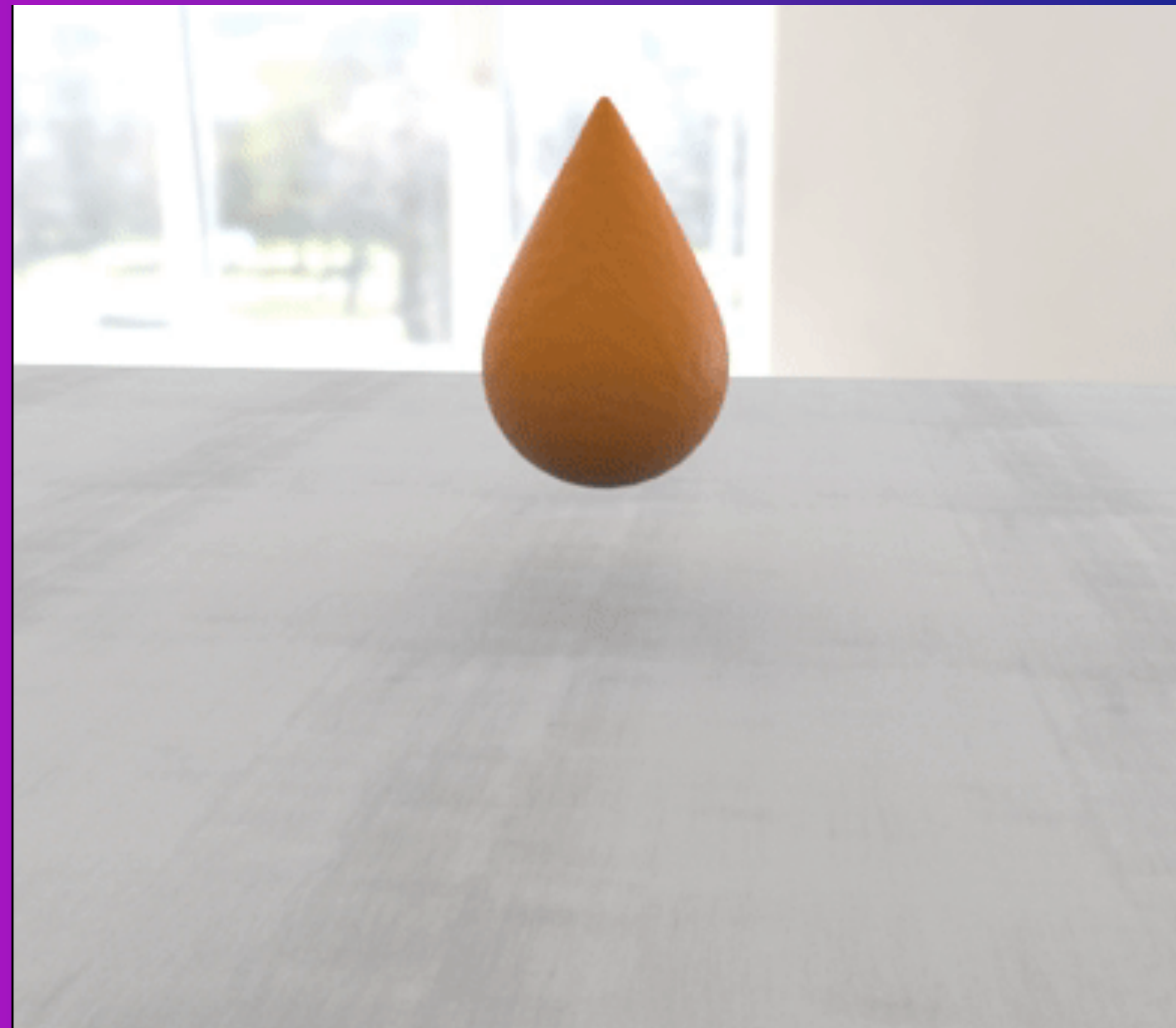


Dynamic Neural Radiance Field

↪ Pumarola et. al. 2021.

↪ Novel spatial and temporal view synthesis

PAC-NeRF



Physics-Augmented Continuum

↪ Li et al. 2023.

↪ Newtonian physics simulations.

↪ Advect via MPM.

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↪ Physics-augmented novel view synthesis for dynamic scenes

$$\frac{D\sigma}{Dt} = \frac{D\mathbf{c}}{Dt} = 0$$

↪ D is material derivative

$$\frac{D\phi}{Dt} = \frac{\partial\phi}{\partial t} + \mathbf{v} \cdot \nabla\phi$$